

# AudioBuddy: Using Sound Sensors to Support Sound Sensitivity Awareness in Autistic Individuals

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## ABSTRACT

More than half of autistic children experience auditory sensitivity and emotion dysregulation. Emotion dysregulation co-occurring with painful or unpleasant experiences due to auditory stimuli can lead to heightened stress, anxiety, and other mental health concerns. Researchers have aimed to support emotion regulation and auditory sensitivity separate from each other. This work builds on those single-purpose efforts to simultaneously address both emotion regulation and auditory sensitivity, with emphasis on self-advocacy for needs related to autism and support for autistic people to take control of their auditory environments themselves. Based on empirical data from two autistic self-advocacy online support forums and existing research, we designed and developed AudioBuddy. AudioBuddy is an assistive multi-device application that notifies users of increased sound levels in their environment while providing coping methods to support emotion regulation and encouraging self-advocacy to change challenging situations.

## CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in accessibility**; **Accessibility technologies**.

## KEYWORDS

Sound Sensors; Smartwatch; Emotion Regulation; Sound Sensitivity

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## 1 INTRODUCTION

More than two-thirds of autistic children exhibit sensitivity to auditory stimuli [3], which can co-occur with physical, neurodevelopmental, and neuropsychiatric conditions like anxiety, depression, and bipolar disorder [5]. Some autistic people struggle to regulate arousal and stress [5], especially with increased noise levels or other distressing stimulation. Autistic people can advocate for themselves, have a full range of emotions to express, and are in part disabled by inhospitable environments that should be adapted to fit their needs [4, 6]. Researchers have explored interventions for auditory sensitivity and emotion regulation (ER), including music therapy, emotion recognition training, mindfulness-based interventions, and cognitive behavior therapy (CBT) [1, 7]. The commercial market has developed assistive technology for autistic people indicating options to support ER and auditory sensitivity. Our ongoing work focuses on wearable assistive devices developed with insight from autistic people to help them self-monitor and advocate for themselves. Here we describe an empirical study and early designs and testing for AudioBuddy.

## 2 EMPIRICAL INPUT INTO DESIGN

We conducted a qualitative analysis of 36 posts and 132 responses from two open forums for autistic people. We aimed to understand (1) how sound sensitivity impacts the lives of autistic people, (2) how they perceive and describe the sounds they are sensitive to that can be automatically detected, and (3) how autistic people view coping mechanisms for sound sensitivity that have been reported in the literature (e.g., music therapy). Detailed analysis of our empirical data is out of scope, but we briefly describe key findings that influenced the development of the AudioBuddy. Reported impacts of sound sensitivity were vast and varied across autistic people and over time, including individual discomfort and decreased mental well-being and disruptions in school, social, and work environments as seen

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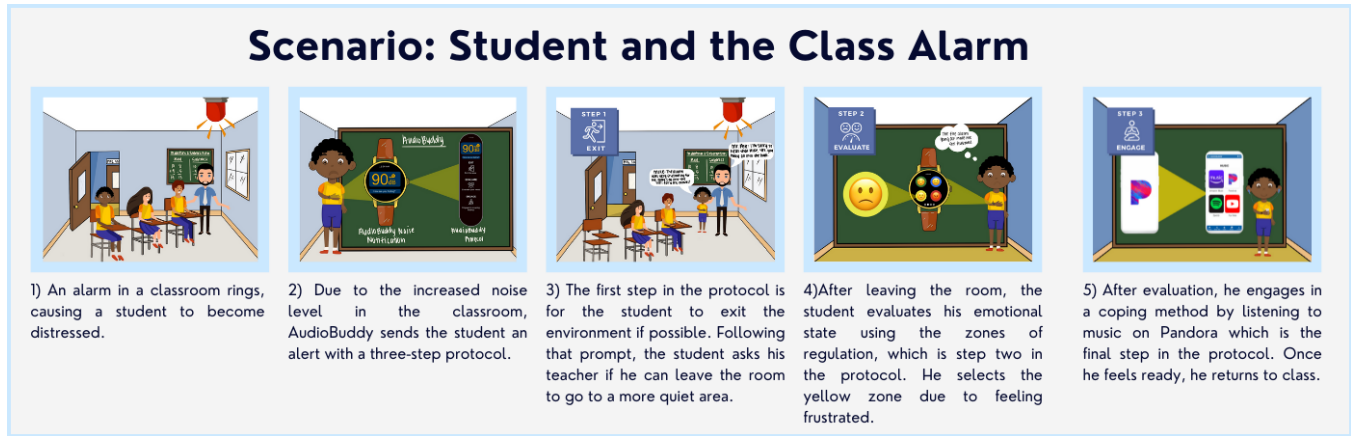


Figure 1: Scenario shows how AudioBuddy could be used in a classroom setting.

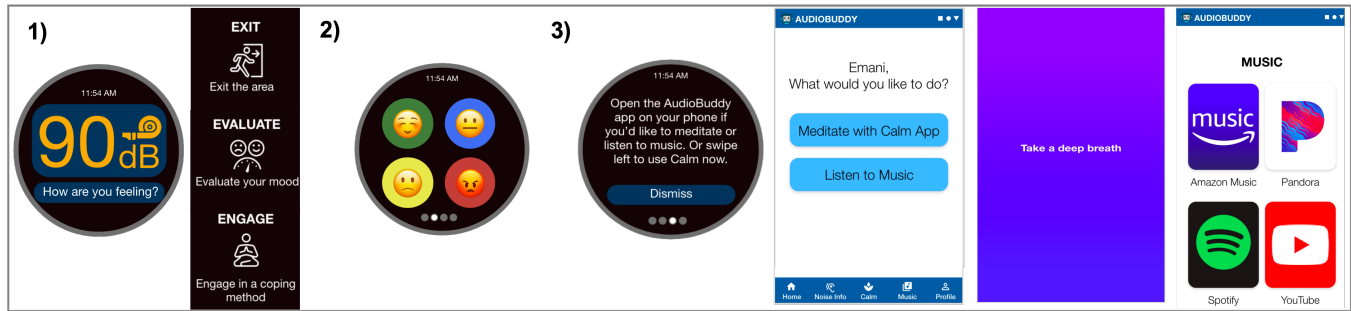


Figure 2: AudioBuddy System. 1) Noise notification with the support protocol. 2) Evaluation through the Zones of Regulation. 3) Engage in a coping strategy to support emotion regulation.

in prior work [1, 3]. Here, we focus on emotion and self-regulation which are essential for coping with environmental disruptions and can quickly become complex. We used a simple regulation process, the zones of regulation (ZOR) by Kuypers [2] which helps individuals understand their feelings, energy, and level of alertness by identifying their zone (e.g., red, yellow, blue, or green zone). For example, someone may identify as being in the blue zone if they are experiencing a low state of alertness and energy, and may feel sad, sick, tired, or bored. The breadth of experiences we saw reported in the forums helps consider ubicomp interventions to support ER and auditory sensitivity simultaneously. For this work, we focus on wearables to allow individuals to work on their self-regulation and provide support for self-advocacy and encouragement of awareness for broader structural change.

### 3 SYSTEM DESIGN AND DEVELOPMENT

AudioBuddy (AB) is an assistive multi-device application that notifies users of increased sound levels in their environment while providing coping methods to support ER and to encourage self-advocacy. AB works as a smartwatch only, smartphone only, or multi-device experience. The system includes self-regulation and emotion control principles from the ZOR [2]. AB measures environmental noise and sends a notification to the user when noise levels

are above a customized threshold. AB then presents options to the user to support self-regulation and encourage changes, which includes reporting their current emotions and alertness, use of third-party apps (e.g., Music, Spotify, Pandora, YouTube, and Calm), and educational materials for self-advocacy. AB can be applicable in settings where the noise level can be unpredictable such as a class room, public transit, restaurants, or places of employment (see Figure 1). Users can also proactively engage with AB outside of loud noise contexts to get support or to reflect on their data.

### 4 USABILITY AND FEASIBILITY TESTING

We conducted two preliminary tests with AB: (1) functionality and power efficiency and (2) user experience with experts and one with an autistic child and his mother. We analyzed battery use for a Samsung Galaxy A51 and Galaxy Watch 4 while sensing ambient noise with different sound levels of sampling (i.e., 5000ms, 50ms, and 1ms). We observed the performance over 4 hours and the results indicated that optimal performance was between 50ms and 500ms which is ideal because this sampling rate will not drain the device's battery. We tested usability in-lab with experts in user experience, two of whom have existing sound sensitivities but none of whom identify as autistic. They iteratively tested the application through multiple rounds of prototyping. Challenges included understanding

the emotion prompts (Figure 2). We changed our design to include three levels of cueing for emotions (color, standard placement on screen, and icon) to correlate with emotions and colors used in the ZOR. Finally, we conducted a preliminary end-user test and one on one semi-structured interview with an autistic child and his mother, who were already familiar with ZOR framework. They both reported the system design to be simple and easy to navigate. They reported their familiarity with ZOR to be helpful, but the mother noted that they thought the design would be understandable even for someone unfamiliar. The mother voiced some concerns including having access to the watch and the phone in a school setting and the potential challenges this may cause.

## 5 CONCLUSIONS AND FUTURE WORK

AudioBuddy, built on the zones of regulation and findings of an empirical study of two autistic online forums, is designed to collectively support autistic people by addressing emotion regulation and auditory sensitivity. Our preliminary evaluation indicates that the noise sampling should be between 50ms and 500ms, the emotion recognition prompt be redesigned for clear understanding, and families can learn about the ZOR prior to using our app. The next step in this work is to collect additional empirical evidence and design feedback from autistic children and adults coping with sound sensitivity and deploy this system to understand its impact and potential for efficacy.

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